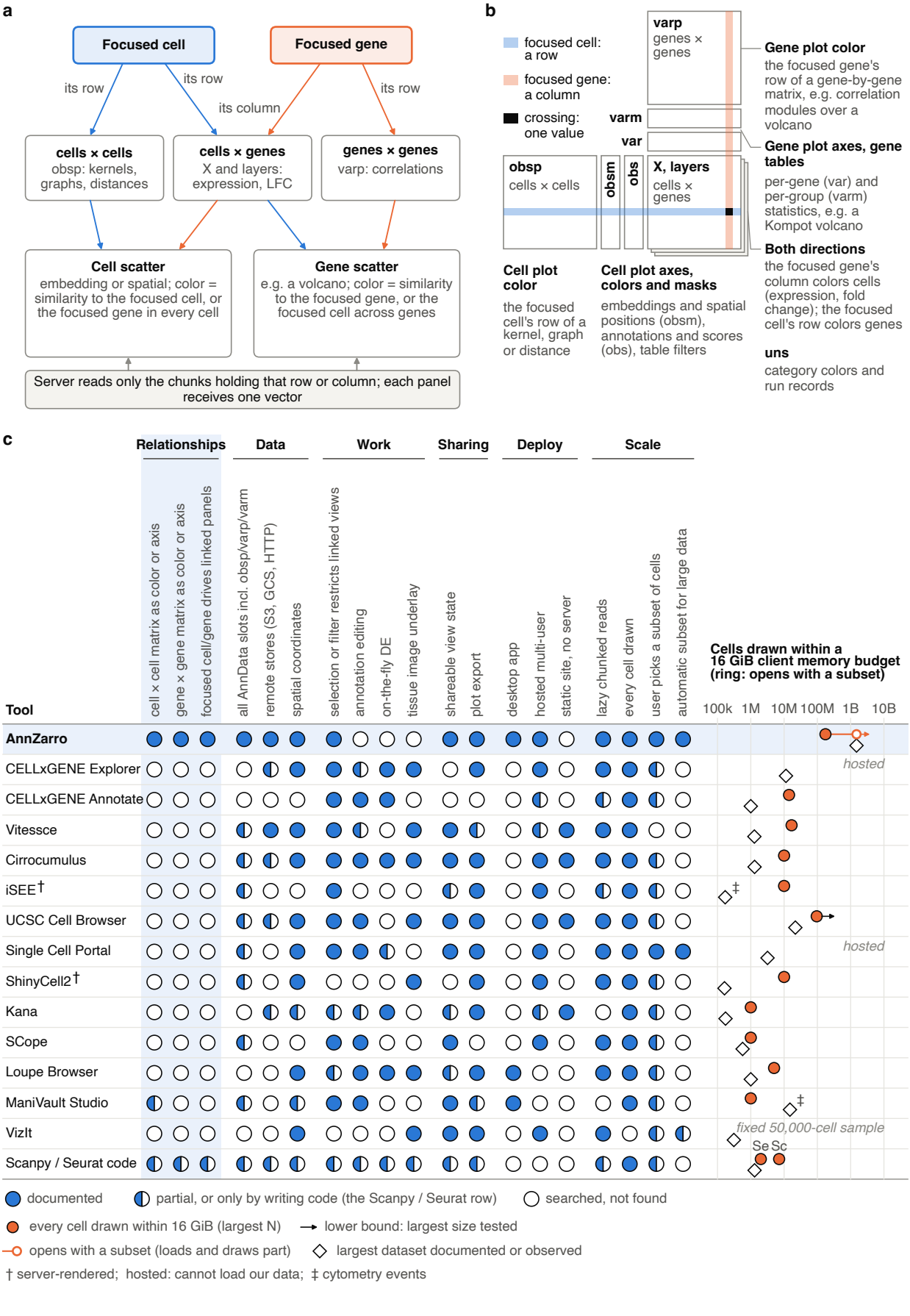

a
Focused cell
Focused gene
its row
its row
its column
its row
cells × cells
obsp: kernels, graphs, distances
cells × genes
X and layers: expression, LFC
genes × genes
varp: correlations
Cell scatter
embedding or spatial; color = similarity to the focused cell, or the focused gene in every cell
Gene scatter
e.g. a volcano; color = similarity to the focused gene, or the focused cell across genes
Server reads only the chunks holding that row or column; each panel receives one vector
b
focused cell: a row
focused gene: a column
crossing: one value
varp
genes × genes
varm
var
obsp
cells × cells
obsm
obs
X, layers
cells × genes
Gene plot color
the focused gene's row of a gene-by-gene matrix, e.g. correlation modules over a volcano
Gene plot axes, gene tables
per-gene (var) and per-group (varm) statistics, e.g. a Kompot volcano
Both directions
the focused gene's column colors cells (expression, fold change); the focused cell's row colors genes
Cell plot color
the focused cell's row of a kernel, graph or distance
Cell plot axes, colors and masks
embeddings and spatial positions (obsm), annotations and scores (obs), table filters
uns
category colors and run records
c
Relationships
Data
Work
Sharing
Deploy
Scale
Tool
cell × cell matrix as color or axis
gene × gene matrix as color or axis
focused cell/gene drives linked panels
all AnnData slots incl. obsp/varp/varm
remote stores (S3, GCS, HTTP)
spatial coordinates
selection or filter restricts linked views
annotation editing
on-the-fly DE
tissue image underlay
shareable view state
plot export
desktop app
hosted multi-user
static site, no server
lazy chunked reads
every cell drawn
user picks a subset of cells
automatic subset for large data
Cells drawn within a 16 GiB client memory budget (ring: opens with a subset)
100k
1M
10M
100M
1B
10B
AnnZarro
CELLxGENE Explorer
hosted
CELLxGENE Annotate
Vitessce
Cirrocumulus
iSEE†
‡
UCSC Cell Browser
Single Cell Portal
hosted
ShinyCell2†
Kana
SCope
Loupe Browser
ManiVault Studio
‡
VizIt
fixed 50,000-cell sample
Scanpy / Seurat code
Se
Sc
documented
partial, or only by writing code (the Scanpy / Seurat row)
searched, not found
every cell drawn within 16 GiB (largest N)
lower bound: largest size tested
opens with a subset (loads and draws part)
largest dataset documented or observed
† server-rendered; hosted: cannot load our data; ‡ cytometry events